



Real-Time Performance and Operational Reliability Monitoring of a Mobile Photovoltaic Power System Using an IoT-Enabled Multilayer Perceptron

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Received: 06-21-2026

Revised: 08-22-2026

Accepted: 08-29-2026

Citation: M. K. Asy'ari, A. Kurniawan, L. J. Mawarani, A. Musyafa, I. Abadi, A. N. Aufa, and A. U. Nu'ila, "Real-time performance and operational reliability monitoring of a mobile photovoltaic power system using an IoT-enabled multilayer perceptron," *Power Eng. Eng. Thermophys.*, vol. 5, no. 3, pp. 234–248, 2026. <https://doi.org/10.56578/peet050305>.



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Abstract: Mobile photovoltaic (PV) power systems provide a flexible electricity supply for remote locations, emergency operations, and other off-grid applications. Their practical operation requires continuous assessment of power conversion performance under changing environmental conditions. This study investigates the efficiency, power output, and operational reliability of a mobile PV system through an Internet of Things (IoT)-enabled monitoring platform and a multilayer perceptron (MLP) model. Solar irradiance, panel temperature, voltage, current, power, and battery state of charge (SOC) were recorded under outdoor operating conditions, yielding approximately 1,600 observations. An MLP with two hidden layers was trained using the Levenberg–Marquardt algorithm, and the data were divided into training and testing subsets at a ratio of 80:20. Operational reliability was evaluated by comparing measured and predicted power outputs against statistically defined control limits. The PV panel achieved an average operating efficiency of approximately 15%, whereas the efficiency of the solar charge controller (SCC) reached 60%. For the normalized dataset, the MLP produced mean squared error (MSE) values of 0.002402 and 0.001951, root mean squared error (RMSE) values of 0.049012 and 0.044173, mean absolute error (MAE) values of 0.033774 and 0.027760, and R^2 values of 0.964491 and 0.970248 for PV and controller power, respectively. The predicted outputs remained within the established control limits throughout the observation period. These findings indicate that the proposed framework can support real-time power-performance assessment and the early identification of abnormal operating conditions in mobile PV systems.

Keywords: Internet of Things; Mobile photovoltaic system; Operational reliability; Power output prediction; Solar charge controller

1 Introduction

Solar energy is a widely available renewable resource that can be converted directly into electricity without the combustion of fossil fuels [1, 2]. This conversion takes place through the photovoltaic (PV) effect in semiconductor solar cells, which constitute the basic power-generating units of a PV system [3, 4]. The electrical output of these cells depends strongly on the solar energy incident on their surfaces. Indonesia has considerable solar energy potential, estimated at approximately 4.8 kWh/m² per day or an equivalent capacity of 112,000 GWp. To support the use of this resource, the Indonesian government has introduced a solar energy development roadmap targeting an installed capacity of 0.87 GW, with an annual addition of approximately 50 MWp [5]. Although utility-scale PV plants generally require substantial installation areas, PV modules can also be assembled into systems of different capacities and configurations for a broad range of power-generation applications [6, 7].

One such configuration is the mobile solar power plant, which combines PV generation, power regulation, and electrical energy storage in a transportable unit. Its portability allows electricity to be supplied at locations that are not connected to the main grid [8, 9]. Mobile PV systems are therefore particularly relevant to remote communities, emergency response activities, temporary field operations, and other situations in which conventional electricity supplies are unavailable or unreliable. Their mobility, however, also exposes them to changing irradiance,

temperature, load, and installation conditions. Consequently, their operating performance cannot be assessed solely from nominal component ratings.

Efficiency and reliability are central concerns in the operation of these systems, particularly when the generated electricity must be transferred through a solar charge controller (SCC) and stored in a battery [10, 11]. Variations in the connected load and battery charging condition can affect the power drawn from the PV panel and the performance of the overall system. Environmental conditions introduce an additional source of variation. Solar irradiance is especially important because an increase in irradiance generally raises the current generated by the PV panel and, consequently, its electrical power output [12–14]. Panel temperature and other operating variables can also alter the relationship between incident solar energy and delivered electrical power. Continuous measurements of these variables are therefore required to distinguish normal operating fluctuations from unusual changes that may require inspection or maintenance. In this context, preventive maintenance supported by operating data can assist in identifying abnormal behaviour before it develops into a serious system failure [15].

Machine learning (ML) methods provide a means of modelling the nonlinear relationships among environmental conditions, electrical variables, and PV power output [16]. Among these methods, the multilayer perceptron (MLP) has been widely used for PV power prediction because it can approximate nonlinear input-output relationships while retaining a relatively simple network structure [17–20]. Previous studies have reported that MLP models can produce competitive results when compared with linear regression, support vector regression, Random Forest, and other conventional prediction methods. Their moderate computational requirements also make them suitable for monitoring applications in which data must be processed repeatedly as operating conditions change [17, 21, 22]. These characteristics are relevant to mobile PV systems, where irradiance, panel temperature, battery condition, and electrical load may vary throughout the operating period.

ML techniques have also been incorporated into PV monitoring and optimization systems to improve power prediction and support the detection of operational anomalies. At the control level, maximum power point tracking methods, including the Incremental Conductance algorithm, are commonly used to locate the PV operating point that provides the highest available power under changing irradiance conditions [23]. At the data-analysis level, comparative studies have shown that Random Forest can outperform a single Decision Tree in PV power prediction. These findings demonstrate the importance of selecting an appropriate prediction method as well as accounting for the operating conditions that govern PV efficiency and power production [24, 25]. Nevertheless, power prediction accuracy alone does not provide a complete assessment of a mobile PV plant. The predicted response must also be related to the measured power-conversion performance and to an operational criterion capable of indicating departures from expected behaviour.

Many existing studies concentrate on either PV power forecasting, maximum power point tracking, or component-level monitoring. Less attention has been given to combining real-time field measurements, PV and SCC efficiency evaluation, power prediction, and control-limit-based reliability indication within a single mobile PV platform. This distinction is important because a mobile system includes not only the PV panel but also the SCC and battery, and losses or irregularities can occur at different stages of the power conversion and storage process. A monitoring framework that follows both PV and SCC outputs can provide a more complete representation of system operation than an assessment based only on panel output.

This study therefore develops a real-time efficiency and operational reliability monitoring system for a mobile PV power plant. The system combines an Internet of Things (IoT)-based data acquisition platform with an MLP power prediction model and a control-limit-based reliability indicator. The work has three main components. First, an IoT monitoring platform was constructed to acquire solar irradiance, temperature, voltage, current, power, and battery state-of-charge data under outdoor operating conditions. Second, MLP models were implemented to predict the output power of the PV panel and SCC under varying operating conditions. Third, the measured and predicted power values were incorporated into an operational reliability assessment based on control limits. In this study, reliability refers to the consistency of observed operating performance and the indication of abnormal behaviour; it does not represent a component lifetime or failure-time estimate. By jointly examining PV conversion efficiency, SCC efficiency, power prediction, and operational consistency, the proposed framework supports continuous condition assessment of mobile PV power systems.

2 Methodology

This study uses an MLP algorithm to design and implement a real-time solar panel reliability monitoring and prediction system based on battery capacity variations.

2.1 Development of Mobile Solar Power Plant

Sensor calibration was carried out prior to the construction of the mobile solar power plant. The calibration included an analysis of measurement uncertainty and the static characteristics of the sensors, as presented in Tables 1–8.

Table 1. Measurement uncertainty of temperature sensor 1

Parameters	Value
UA_1	0.348 °C
UA_2	0.575 °C
UB_1	0.0288 °C
UB_2	0.288 °C
UC	0.732 °C
V_{eff}	7.53
k	2.376
U_{exp}	1.706 °C
CL	98%

Note: UA_1 = uncertainty from scale measurement; UA_2 = uncertainty from repeated measurements; UB_1 = calibrator resolution uncertainty; UB_2 = calibrator uncertainty; UC = combined standard uncertainty; V_{eff} = effective degrees of freedom; k = coverage factor; U_{exp} = expanded uncertainty; CL = confidence level.

Table 2. Static characteristics of temperature sensor 1

Parameters	Value
Accuracy	1.99%
Precision	98.61%
Linierity	0.45%
Hysteresis	0.182%

Table 3. Measurement uncertainty of temperature sensor 2

Parameters	Value
UA_1	0.016 °C
UA_2	0.036 °C
UB_1	0.02886 °C
UB_2	0.288 °C
UC	0.29277 °C
V_{eff}	13,359.62
k	2.376
U_{exp}	0.696 °C
CL	98%

Note: UA_1 = uncertainty from scale measurement; UA_2 = uncertainty from repeated measurements; UB_1 = calibrator resolution uncertainty; UB_2 = calibrator uncertainty; UC = combined standard uncertainty; V_{eff} = effective degrees of freedom; k = coverage factor; U_{exp} = expanded uncertainty; CL = confidence level.

Table 4. Static characteristics of temperature sensor 2

Parameters	Value
Accuracy	1.44%
Precision	98.99%
Linierity	0.512%
Hysteresis	0.412%

The mobile solar power plant was developed to establish a monitoring system capable of accurately capturing key operational parameters affecting system performance. This process involves identifying critical parameters that influence solar irradiation, selecting appropriate sensors, and formulating detailed technical specifications for system components, including sensors, hardware, and software. The design scheme for the mobile solar power plant can be seen in Figure 1, and the wiring architecture for monitoring the efficiency and reliability of solar panels can be seen in Figure 2.

The monitoring system acquires solar irradiance, temperature, voltage, current, and power data from the PV panel and SCC. The collected data are processed by the ESP32 microcontroller and transmitted to a cloud platform for storage and further analysis. The monitoring results can be accessed remotely through internet-connected devices.

Table 5. Measurement uncertainty of PZEM-017 sensor 1

Parameters	Value
UA_1	0.02 A
UA_2	0.054 A
UB_1	0.0288 A
UB_2	0.5773 A
UC	0.58 A
V_{eff}	38,903.65
k	2.376
U_{exp}	1.38 A
CL	98%

Note: PZEM-017 = direct current (DC) power and energy measurement module; UA_1 = uncertainty from scale measurement; UA_2 = uncertainty from repeated measurements; UB_1 = calibrator resolution uncertainty; UB_2 = calibrator uncertainty; UC = combined standard uncertainty; V_{eff} = effective degrees of freedom; k = coverage factor; U_{exp} = expanded uncertainty; CL = confidence level.

Table 6. Static characteristics of PZEM-017 sensor 1

Parameters	Value
Accuracy	1.91%
Precision	98.21%
Linierity	0.489%
Hysteresis	0.14%

Note: PZEM-017 = direct current (DC) power and energy measurement module.

Table 7. Measurement uncertainty of PZEM-017 sensor 2

Parameters	Value
UA_1	0.02 A
UA_2	0.054 A
UB_1	0.0288 A
UB_2	0.5773 A
UC	0.58 A
V_{eff}	38,903.65
k	2.376
U_{exp}	1.38 A
CL	98%

Note: PZEM-017 = direct current (DC) power and energy measurement module; UA_1 = uncertainty from scale measurement; UA_2 = uncertainty from repeated measurements; UB_1 = calibrator resolution uncertainty; UB_2 = calibrator uncertainty; UC = combined standard uncertainty; V_{eff} = effective degrees of freedom; k = coverage factor; U_{exp} = expanded uncertainty; CL = confidence level.

Table 8. Static characteristics of PZEM-017 sensor 2

Parameters	Value
Accuracy	1.96%
Precision	98.98%
Linierity	0.034%
Hysteresis	0.5%

Note: PZEM-017 = direct current (DC) power and energy measurement module.

2.2 Monitoring Efficiency

Efficiency monitoring uses solar irradiance, temperature, and humidity as inputs, while voltage, current, and power are used as outputs. The configuration table of the solar panel monitoring system can be seen in Table 9 below.

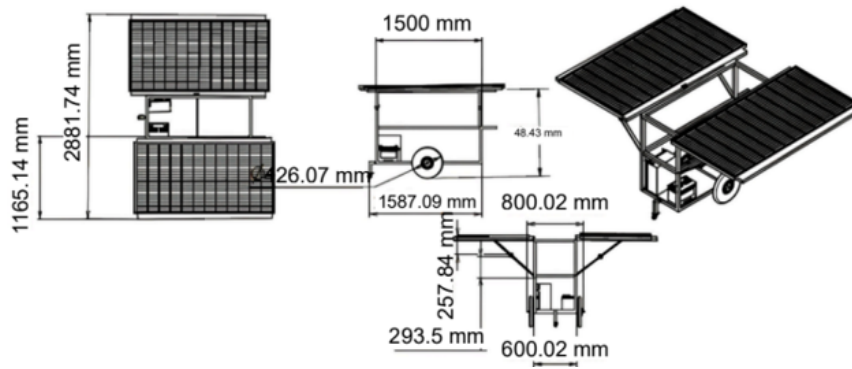


Figure 1. 3D mobile solar power plant

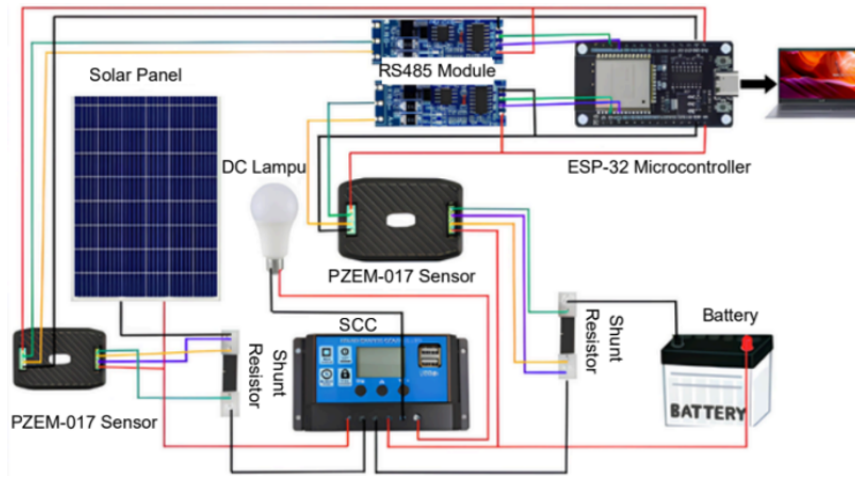


Figure 2. Architecture of solar panel efficiency and reliability monitoring system

Note: RS485 = Recommended Standard 485; DC = direct current; SCC = solar charge controller; ESP32 = ESP32 microcontroller; PZEM-017 = DC power and energy measurement module. PZEM-017 and ESP32 are device/model designations.

Table 9. Wiring configuration of solar panel monitoring system

Component	Power Supply		Input A/D	Output A/D
	(+)	(-)		
ESP32	Supply DC (12V) +	Supply DC (12V) -	GPIO26, GPIO27, GPIO5, GPIO4, RX2, TX2, RX0, TX0	Solar irradiance, temperature, current, voltage, power
Solar irradiance sensor	Supply DC Battery (12V) +	Supply DC Battery (12V) -	Solar irradiance	GPIO26, GPIO27
DS18B20 (1)	ESP32 Vcc (+)	ESP32 Gnd (+)	Temperature	GPIO4
DS18B20 (2)	ESP32 Vcc (+)	ESP32 Gnd (+)	Temperature	GPIO5
PZEM-017 (1)	ESP32 Vcc (+)	ESP32 Vcc (+)	Output Panel Surya	RX2, TX2
PZEM-017 (2)	ESP32 Vcc (+)	ESP32 Vcc (+)	Output SCC	RX0, TX0

Note: ESP32 = ESP32 microcontroller; DC = direct current; A/D = analog/digital; GPIO = general-purpose input/output; RX = receive; TX = transmit; Vcc = supply voltage; Gnd = ground; SCC = solar charge controller; DS18B20 = digital temperature sensor; PZEM-017 = DC power and energy measurement module; SCC = solar charge controller.

Calculation of solar panel efficiency using Eqs. (1) and (2).

$$\eta_{PV} = \frac{P_{PV}}{I_r \times A_{PV}} \times 100\% \quad (1)$$

$$\eta_{SCC} = \frac{P_{SCC}}{P_{PV}} \times 100\% \quad (2)$$

where, η is the efficiency, P_{PV} is the output power of the solar panel, P_{SCC} is the output power of the SCC, I_r is the solar irradiation, and A_{PV} is the area of the solar panel. The reported efficiency values represent average operational efficiencies obtained during the observation period.

2.2.1 Designing an IoT system

The ThingSpeak IoT data transmission flow diagram can be seen in Figure 3 below.

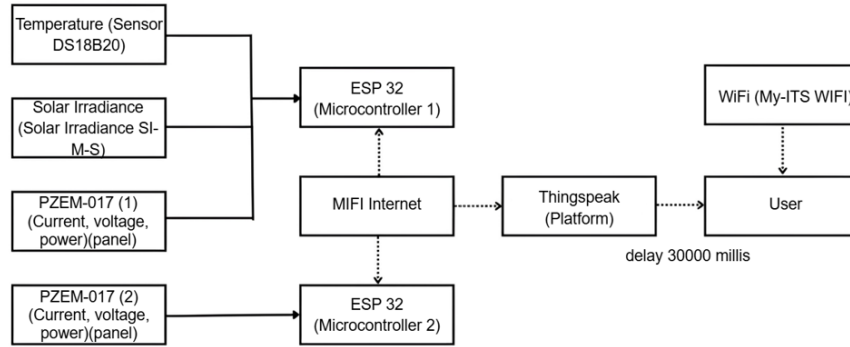


Figure 3. ThingSpeak Internet of Things (IoT) data transmission flowchart

Note: ESP32 = ESP32 microcontroller; PZEM-017 = direct current (DC) power and energy measurement module; DS18B20 = digital temperature sensor; SI-M-S = solar irradiance sensor model; MiFi = mobile Wi-Fi hotspot; Wi-Fi = wireless network technology.

The IoT system design was developed using the ESP32 microcontroller and the ThingSpeak cloud platform to enable real-time data acquisition, storage, and visualization. Sensor data, including solar irradiance, temperature, voltage, current, and power, were transmitted to the cloud platform and subsequently used for efficiency analysis, power prediction, and operational reliability monitoring.

2.2.2 Designing a multilayer perceptron model

The MLP was implemented in MATLAB using the Neural Network Toolbox. The model received three input variables and comprised two hidden layers, each containing 10 neurons, followed by a single output neuron. A tangent sigmoid function was applied to the hidden layers, whereas a linear function was assigned to the output layer. Prior to model training, all input and output data were scaled to a 0–1 range. The available dataset was subsequently separated into training and testing sets using an 80:20 proportion. Model optimization was conducted using the Levenberg–Marquardt backpropagation algorithm, with the training process limited to a maximum of 1000 epochs [26, 27]. The trained MLP was then assessed based on its capability to represent the nonlinear relationship between the selected input variables and the target output. Five statistical measures were employed to quantify the predictive performance, namely mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and the coefficient of determination (R^2). Together, these indicators provide complementary information regarding the accuracy and consistency of the model predictions. MSE quantifies the mean of the squared residuals, thereby emphasizing larger prediction errors. RMSE is obtained from the square root of MSE and expresses the prediction error in the same unit as the predicted variable. In contrast, MAE represents the average magnitude of the absolute prediction errors without considering their direction. MAPE evaluates the average prediction error relative to the corresponding actual values and expresses the result as a percentage. Meanwhile, R^2 indicates the fraction of the variation in the target variable accounted for by the MLP model; a value approaching 1 indicates a strong agreement between the model predictions and the observed data [26, 28].

2.2.3 Data acquisition procedure

Data were collected continuous from 06:00 to 17:00 local time under outdoor operating conditions. The monitoring system recorded solar irradiance, temperature, voltage, current, power, and battery state of charge (SOC) at regular intervals throughout the observation period. A total of approximately 1,600 observations were acquired and subsequently used for training and testing MLP model. The collected dataset represents the actual operating behaviour of the mobile solar power plant under varying environmental conditions.

2.3 Operational Monitoring Reliability

In this study, operational reliability monitoring was conducted using reliability indicators for the PV panel (R_{PV}) and SCC (R_{SCC}). Reliability was evaluated based on the consistency of the system's operating performance by comparing actual and predicted power outputs. A failure event was defined as an observation exceeding the established control limits, indicating a deviation from the expected operating behaviour. The reliability indicator was

subsequently calculated using the failure function and Benard's approximation. Therefore, the obtained reliability values represent indicators of operational performance consistency and early abnormality detection rather than component lifetime reliability. Reliability calculations are based on Benard's approximation in Eq. (3) [29].

$$R(i) = 1 - \frac{D(i) - 0.3}{N_D + 0.4} \quad (3)$$

where, (i) is the current reliability in i , $D(i)$ is the cumulative failure, and N_D is the total number of failures.

3 Results

The results and discussion of the research conducted based on the methods described above are as follows.

3.1 Efficiency Monitoring Result

Solar panel efficiency measures how effectively the panel converts solar irradiance into electrical energy, while SCC efficiency represents how efficiently the generated energy is transferred to the battery. The efficiency monitoring results can be seen in Figure 4.

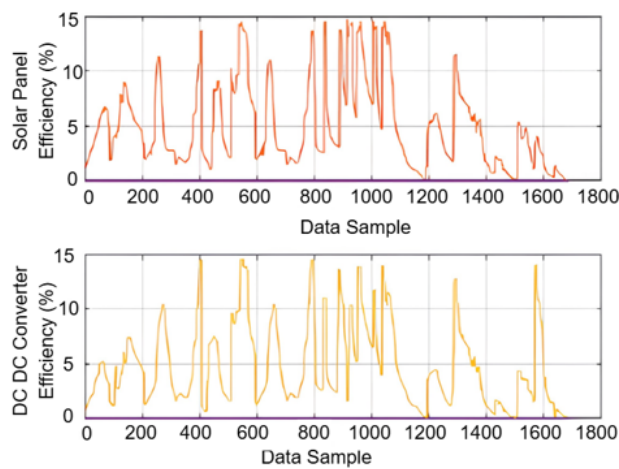


Figure 4. Solar power plant system efficiency graph

Note: DC = direct current.

Based on Figure 4 and Table 10, the solar panel efficiency shows noticeable fluctuations, with values reaching up to approximately 15%, which reflect variations in solar irradiance during the observation period. In contrast, the SCC efficiency is relatively more stable and remains below 60%, indicating that the SCC can regulate and transfer power from the solar panel to the battery in a consistent manner, despite fluctuations in the solar panel output. Panel efficiency refers to the ratio between output electrical power and incident solar energy, whereas SCC efficiency represents the ratio between SCC output power and PV output power averaged over the observation period.

Table 10. Solar power plant efficiency results

Component	Efficiency
Solar panel	15%
SCC	60%

Note: SCC = solar charge controller.

3.1.1 Multilayer perceptron modelling results

The collected data was divided into training data and test data with a composition of 80:20. The solar irradiance intensity ranges from 11.24 to 865.23 Watts/m², while the temperature of the solar panel ranges from 24 to 53.64 °C. The panel output voltage ranges from 51.17 to 75.64 Volts, the panel output current ranges from 0.03 to 5.73 Amperes, and the panel output power ranges from 1.19 to 550.65 Watts. Then, the SCC output voltage ranges from 45.59 to 62.9 volts, the SCC output current ranges from 0.03 to 5.73 amperes, and the SCC output power ranges from 1.45 to 323.3 watts. The input and output data exhibit a daily pattern resembling a sinusoidal wave, as shown in Figure 5.

Figure 6 shows a boxplot representing the distribution of data from the input and output variables. The boxplot visualization shows that no data points fall outside the whiskers.

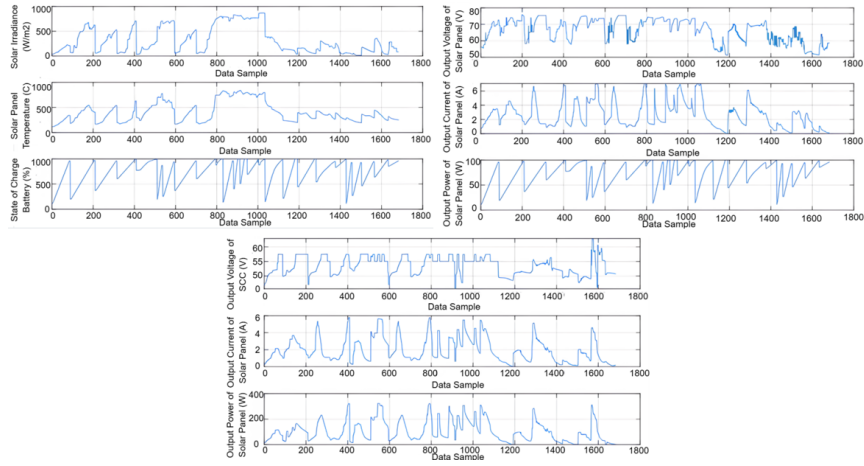


Figure 5. Graph of input and output data

Note: SCC = solar charge controller.

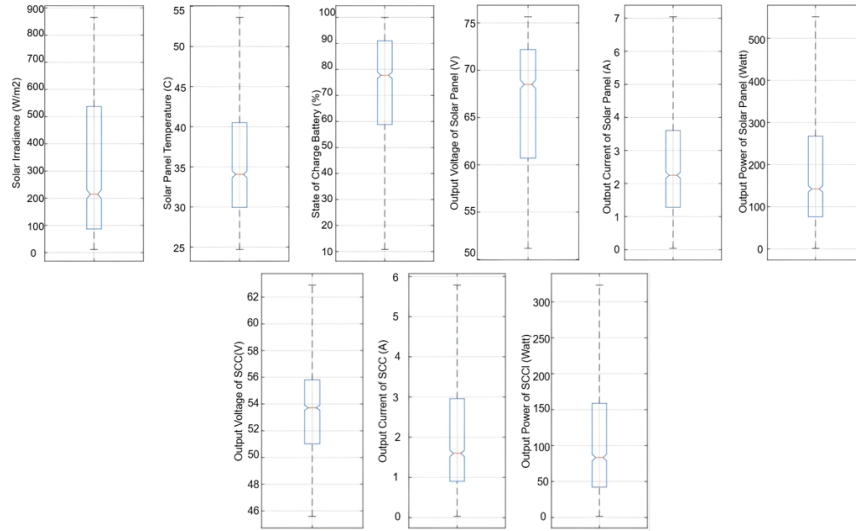


Figure 6. Boxplot of input and output data

Note: SCC = solar charge controller.

Figure 7 shows the histogram distribution of data from the input and output variables of the solar panel. Most variables have a right-skewed distribution pattern.

Figure 8 shows the correlation coefficients among the measured parameters of the mobile solar power plant. The results indicate a strong positive association between solar irradiance and panel temperature (0.89), suggesting that higher irradiance levels are generally accompanied by increased panel temperatures during operation. This observation is consistent with previous findings [28]. Panel power output exhibits a strong correlation with panel current (0.92), while SCC power output shows a similarly strong correlation with SCC current (0.93). In contrast, the battery SOC demonstrates relatively weak correlations with most variables, indicating that its behavior is influenced by multiple operating factors beyond the instantaneous environmental and electrical parameters considered in this study. It should be noted that correlation analysis reflects statistical associations between variables and should not be interpreted as evidence of direct causal relationships.

Figure 9 shows a normalized data graph from training and testing data. Data normalization uses a range of 0 to 1.

Figure 10 illustrates the architecture of the MLP model, which consists of three input variables, two hidden layers with 10 neurons each, and one output neuron.

Figures 11 and 12 show the comparison between actual and predicted values during the training and testing processes. The predicted curves closely match the actual power variations for both input and output power, indicating that the MLP model successfully learns the underlying relationship between the measured parameters and power generation.

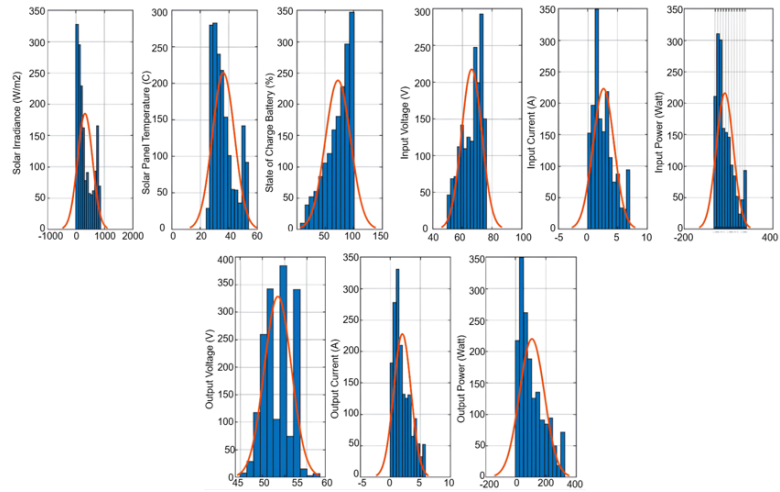


Figure 7. Histogram of input and output data

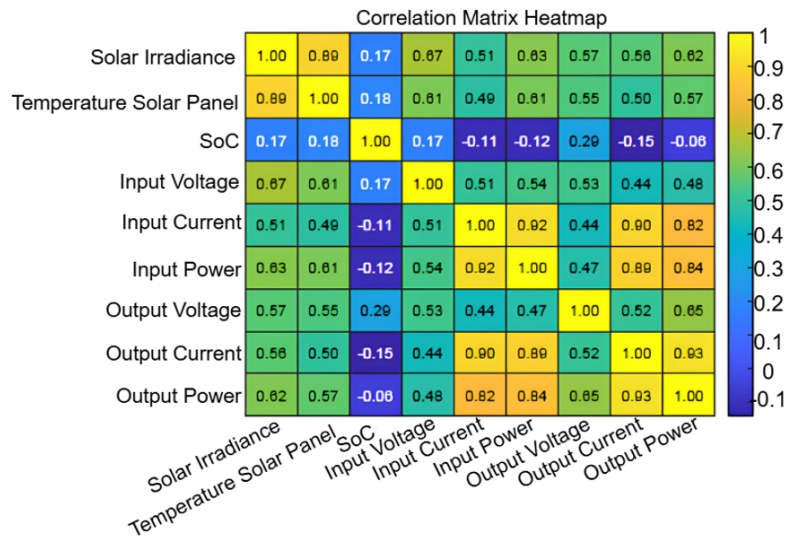


Figure 8. Correlation matrix

Note: SOC = state of charge.

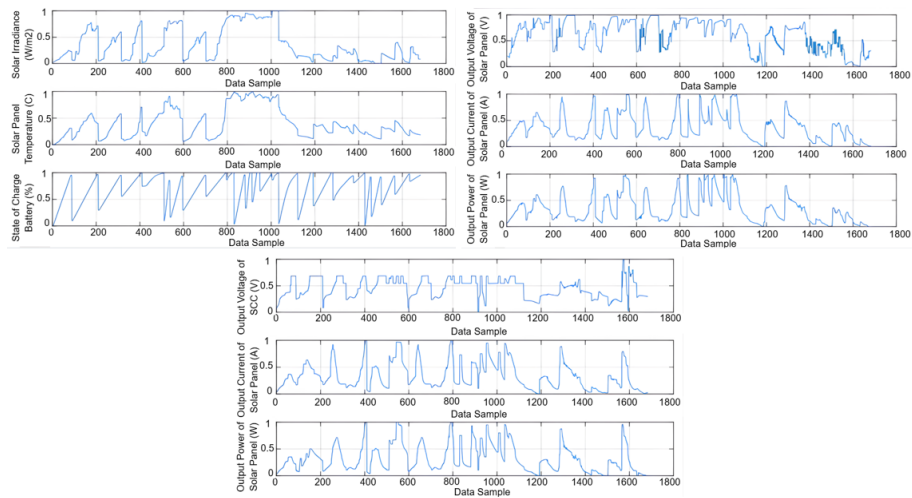


Figure 9. Normalized data graph

Note: SCC = solar charge controller.

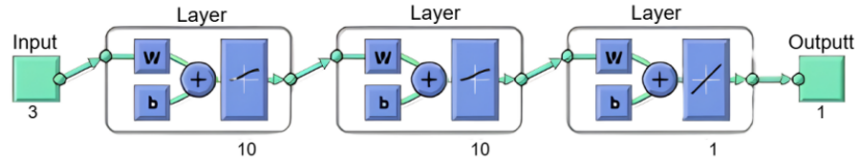


Figure 10. Multilayer perceptron (MLP) architecture

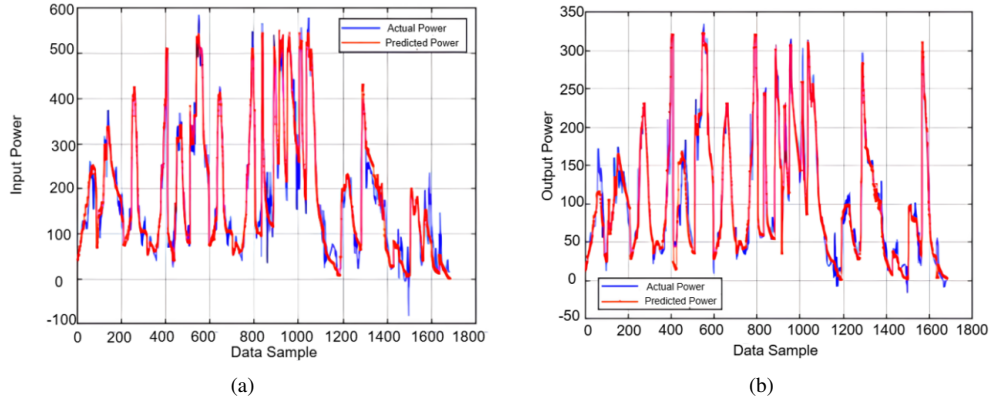


Figure 11. Graph of actual and predicted values during the training process for: (a) solar panel input power; (b) solar panel output power

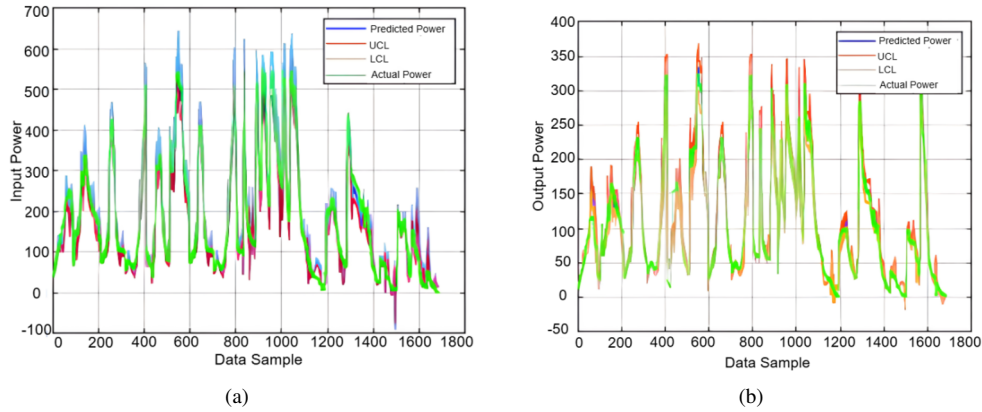


Figure 12. Graph of actual and predicted values during the testing process for: (a) solar panel input power; (b) solar panel output power

Note: UCL = upper control limits; LCL = lower control limits.

RMSE, MAE, MAPE, and R^2 were used because they are widely adopted evaluation metrics for PV power forecasting and provide complementary information on prediction accuracy and model generalization capability [30]. The evaluation metrics were calculated using normalized dataset to ensure consistency with the training process and to provide a fair assessment of the model performance. Table 11 shows that the proposed MLP model achieved high prediction accuracy for both PV power and SCC power outputs.

Table 11. Evaluation metrics result

Output	MSE	RMSE	MAE	R^2
PV power	0.002402	0.049012	0.033774	0.964491
SCC power	0.001951	0.044173	0.027760	0.970248

Note: MSE = mean squared error; RMSE = root mean squared error; MAE = mean absolute error; PV = photovoltaic; SCC = solar charge controller; R^2 = coefficient of determination.

3.2 Operational Reliability Monitoring Result

Figure 13 shows the reliability analysis results of the solar panel and the SCC based on the comparison of actual power data, predicted power, and control limits.

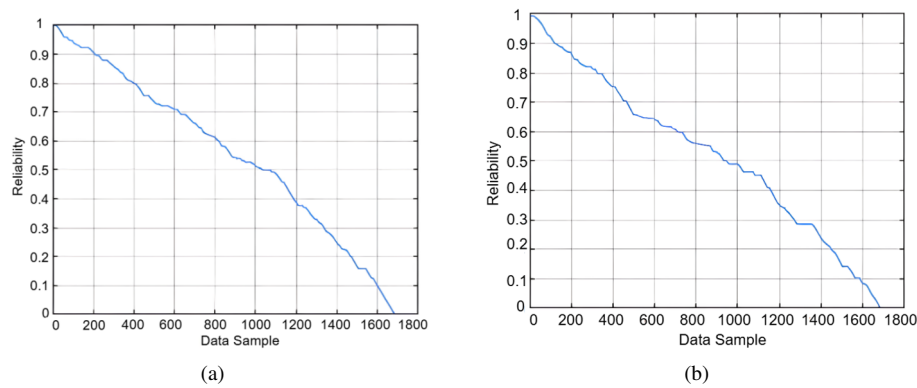


Figure 13. Graphic analysis of reliability: (a) solar panel; (b) solar charge controller (SCC)

The results show that the predicted power remains within the defined control limits. The control limits were established using the upper control limits (UCL) and lower control limits (LCL) obtained from statistical characteristics. The reliability curves show a gradual decrease as the number of data samples increases.

4 Discussion

4.1 Efficiency Analysis

The PV panel exhibited noticeable efficiency variations during the observation period. These variations coincided with changes in solar irradiance and panel temperature, both of which affect the electrical response of PV modules under outdoor conditions. Solar irradiance determines the incident energy available for conversion, whereas an increase in panel temperature may alter the voltage and conversion performance of the module. The observed efficiency pattern should therefore be understood as the combined response of the PV panel to changing environmental and operating conditions rather than as the effect of irradiance alone.

The SCC showed a less variable efficiency pattern than the PV panel, although its efficiency remained below 60% during the measurements. This result indicates that the power transferred from the PV panel was regulated with comparatively limited temporal variation, but it also points to appreciable losses between the PV and SCC outputs. Such losses may be associated with the operating point of the PV panel, the charging state of the battery, the connected load, wiring losses, or the conversion characteristics of the SCC. The current measurements do not allow the contribution of each factor to be separated. Nevertheless, monitoring both PV and SCC efficiency makes it possible to distinguish fluctuations at the power-generation stage from those occurring during power regulation and battery charging.

4.2 Data Characteristics and Statistical Behaviour

The input and output variables followed a daily pattern broadly consistent with the variation in solar irradiance between 06:00 and 17:00. The corresponding changes in panel temperature, current, and power show that the dataset captured the response of the mobile PV system under changing outdoor conditions. This pattern is particularly relevant to power prediction because it includes periods of low, intermediate, and high solar input within the monitored operating window.

No observations appeared beyond the whiskers in the boxplots. This finding suggests that the recorded variables did not contain extreme values according to the adopted boxplot criterion, although it should not be interpreted as proof that all measurements were free from error. The right-skewed distributions further indicate that low-to-medium operating values occurred more frequently than values near the upper end of the measured ranges. Consequently, the model was trained with fewer observations representing peak operating conditions. This imbalance should be considered when interpreting its predictive performance, particularly during periods of high irradiance or high power output.

4.3 Correlation Analysis

The correlation analysis identified several strong statistical associations among the measured variables. Solar irradiance and panel temperature had a correlation coefficient of 0.89, showing that higher irradiance was generally

accompanied by a higher panel temperature during the observation period. Panel power and panel current were also strongly correlated, with a coefficient of 0.92. A similar relationship was found between SCC power and SCC current, for which the correlation coefficient reached 0.93. These results are consistent with the electrical behaviour of the system because variations in current contributed directly to the measured power output.

Battery SOC, in contrast, had relatively weak correlations with most of the instantaneous measurements. This difference is reasonable because SOC reflects the accumulated balance between charging and discharging rather than the operating condition at a single measurement point. Its value may also depend on the previous battery condition, charging duration, load demand, and battery characteristics. The correlation results therefore describe statistical associations within the collected dataset and do not establish causal relationships among the variables.

4.4 Multilayer Perceptron Model Performance

The MLP models reproduced the measured PV and SCC power patterns with relatively small errors on the normalized test data. For PV power prediction, the model produced an MSE of 0.002402, an RMSE of 0.049012, an MAE of 0.033774, and an R^2 of 0.964491. The corresponding values for SCC power prediction were 0.001951, 0.044173, 0.027760, and 0.970248, respectively. Because the error metrics were calculated after normalization, they represent errors on the normalized scale and should not be interpreted directly in watts.

The SCC model showed slightly lower error values and a higher R^2 than the PV model. This difference may be related to the comparatively stable SCC output and the strong association between SCC current and power. The PV output was exposed more directly to fluctuations in irradiance and panel temperature, which may have made its short-term behaviour more difficult to reproduce.

The R^2 values indicate that the two models accounted for more than 96% of the variation observed in their respective test subsets. The close agreement between measured and predicted power demonstrates that the selected MLP architecture learned the principal input–output patterns contained in the present dataset. These findings support its use as a prediction component in the proposed monitoring platform. However, the results are limited to the environmental conditions, system configuration, and observation period represented by the collected data. Measurements obtained across different seasons, locations, loads, and battery conditions would be required before broader predictive capability could be established.

4.5 Operational Reliability Analysis

The operational reliability analysis compared the measured and predicted power values with the established upper and LCL. The predicted PV and SCC power values remained within these limits during the observation period, indicating that the model did not identify a power deviation exceeding the adopted statistical thresholds. Within the scope of the present study, this result represents consistency with the expected operating pattern rather than evidence of component lifetime reliability.

The downward shape of the calculated reliability curves should not be interpreted as proof of gradual physical degradation. The curves were derived from the adopted failure-ranking procedure and the cumulative treatment of observations; therefore, their form is partly determined by the calculation method. In addition, the data were collected over a limited observation period and do not provide the long-term operating history required to quantify ageing or deterioration of the PV panel, SCC, or battery.

The proposed framework is consequently more appropriately described as an operational reliability indicator or an abnormality-screening method. Observations outside the control limits can be used to identify periods that require further examination, but they cannot by themselves determine the cause of an abnormality or predict the remaining lifetime of a component. Longer monitoring periods, confirmed fault records, and repeated measurements under different operating conditions are needed to establish warning thresholds and evaluate the framework for maintenance decision-making.

5 Conclusions

This study developed an IoT-enabled monitoring framework for evaluating the efficiency, power output, and operational reliability of a mobile PV power system. During the observation period, the PV panel had an average operating efficiency of approximately 15%, with variations occurring alongside changes in solar irradiance, panel temperature, and other operating conditions. The SCC exhibited a less variable efficiency pattern, although its efficiency reached only 60%, indicating that losses remained between the PV and SCC outputs.

The MLP models reproduced the measured PV and SCC power patterns with MSE values of 0.002402 and 0.001951, RMSE values of 0.049012 and 0.044173, MAE values of 0.033774 and 0.027760, and R^2 values of 0.964491 and 0.970248, respectively. The error values were calculated using normalized data. These results show that the models captured the principal relationships represented in the collected dataset, with the SCC power model producing slightly lower prediction errors than the PV power model.

The predicted power values remained within the established control limits throughout the monitored period. This finding indicates consistency with the expected operating pattern under the conditions examined, rather than long-term component reliability or physical degradation. The proposed framework provides a practical method for combining real-time measurements, power prediction, efficiency assessment, and control-limit-based abnormality screening in a mobile PV system. Further measurements covering different seasons, locations, load conditions, and confirmed fault events are required to evaluate its performance beyond the present dataset and to establish reliable thresholds for maintenance decisions.

Author Contributions

Conceptualization, M.K.A. and A.K.; methodology, A.K. and A.M.; software, A.K.; validation, I.A. and L.J.M.; formal analysis, A.U.N. and A.N.A.; investigation, A.K.; resources, M.K.A.; data curation, L.J.M.; writing—original draft preparation, A.U.N.; writing—review and editing, M.K.A.; visualization, A.U.N.; supervision, I.A. and A.M.; project administration, L.J.M.; funding acquisition, L.J.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Acknowledgements

The authors gratefully acknowledge Institut Teknologi Sepuluh Nopember for its financial support, as well as for providing the laboratory facilities and research environment necessary to carry out this study.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the authors employed generative AI tools (ChatGPT) and Grammarly solely for language editing and grammatical enhancement. All scientific reasoning, technical development, experimental design, data analysis, and interpretation were performed entirely by the authors. The manuscript was carefully reviewed to ensure scientific accuracy and integrity, and the authors take full responsibility for its content.

References

- [1] M. A. Koondhar, L. Albasha, I. Mahariq, B. B. Graba, and E. Touti, “Reviewing floating photovoltaic (FPV) technology for solar energy generation,” *Energy Strateg. Rev.*, vol. 54, p. 101449, 2024. <https://doi.org/10.1016/j.esr.2024.101449>
- [2] A. Saxena, R. Kumar, A. A. Sagade, D. B. Singh, V. V. Tyagi, E. Cuce, and V. Goel, “A state-of-art review on photovoltaic systems: Design, performance, and progress,” *Process Saf. Environ. Prot.*, vol. 190, no. Part A, pp. 1324–1354, 2024. <https://doi.org/10.1016/j.psep.2024.07.111>
- [3] A. S. Al-ezzi and M. N. M. Ansari, “Photovoltaic solar cells: A review,” *Appl. Syst. Innov.*, vol. 5, no. 4, p. 67, 2022. <https://doi.org/10.3390/asi5040067>
- [4] S. Al-Ali, A. G. Olabi, and M. Mahmoud, “A review of solar photovoltaic technologies: Developments, challenges, and future perspectives,” *Energy Convers. Manag. X*, vol. 27, p. 101057, 2025. <https://doi.org/10.1016/j.ecmx.2025.101057>
- [5] K. T. N. Ihsan, T. S. Anggraini, M. Adrian, P. Rohayani, and A. D. Sakti, “Spatial modeling of multi-scenario optimal solar PV power plant distribution to support Indonesia’s clean energy achievement targets,” *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, vol. XLVI-M-2-2022, pp. 119–126, 2022. <https://doi.org/10.5194/isprs-archives-XLVI-M-2-2022-119-2022>
- [6] A. Allouhi, S. Rehman, M. S. Buker, and Z. Said, “Recent technical approaches for improving energy efficiency and sustainability of PV and PV-T systems: A comprehensive review,” *Sustain. Energy Technol. Assessments*, vol. 56, p. 103026, 2023. <https://doi.org/10.1016/j.seta.2023.103026>
- [7] C. Muensuksaeng, C. Harnmanasvate, J. Chantana, and R. Cheacharoen, “Portable solar-powered dual storage integrated system: A versatile solution for emergency,” *Sol. Energy*, vol. 247, pp. 245–254, 2022. <https://doi.org/10.1016/j.solener.2022.10.030>
- [8] A. Sedaghat, R. Kalbasi, R. Narayanan, A. Mehdizadeh, S. M. Soleimani, M. A. Malayer, M. I. Al-Khiami, H. Salem, W. K. Hussam, M. Sabati, and et al., “Integrating solar PV systems for energy efficiency in portable cabins: A case study in Kuwait,” *Sol. Energy*, vol. 277, p. 112715, 2024. <https://doi.org/10.1016/j.solener.2024.112715>

- [9] T. Li, T. Zhang, and W. Li, "Configuration optimization of mobile photovoltaic-diesel-storage microgrid system based on CPS-MOEA," *Appl. Sci.*, vol. 14, no. 7, p. 3015, 2024. <https://doi.org/10.3390/app14073015>
- [10] F. Schmid and F. Behrendt, "Optimal sizing of solar home systems: Charge controller technology and its influence on system design," *Sustain. Energy Technol. Assessments*, vol. 45, p. 101198, 2021. <https://doi.org/10.1016/j.seta.2021.101198>
- [11] M. Hasan and H. S. Altinoluk, "Current and future prospective for battery controllers of solar PV integrated battery energy storage systems," *Front. Energy Res.*, vol. 11, p. 1139255, 2023. <https://doi.org/10.3389/fenrg.2023.1139255>
- [12] A. Gholami, M. Ameri, M. Zandi, R. G. Ghochani, and H. A. Kazem, "Predicting solar photovoltaic electrical output under variable environmental conditions: Modified semi-empirical correlations for dust," *Energy Sustain. Dev.*, vol. 71, pp. 389–405, 2022. <https://doi.org/10.1016/j.esd.2022.10.012>
- [13] O. Bamişile, C. Acen, D. Cai, Q. Huang, and I. Staffell, "The environmental factors affecting solar photovoltaic output," *Renew. Sustain. Energy Rev.*, vol. 208, p. 115073, 2025. <https://doi.org/10.1016/j.rser.2024.115073>
- [14] E. H. Fatih and S. R. Aslan, "Effects of dust deposition on the surface temperature and performance of photovoltaic modules," *Power Eng. Eng. Thermophys.*, vol. 5, no. 3, pp. 202–208, 2026. <https://doi.org/10.56578/peet050302>
- [15] B. Chen, Y. Liu, C. Zhang, and Z. Wang, "Time series data for equipment reliability analysis with deep learning," *IEEE Access*, vol. 8, pp. 105 484–105 493, 2020. <https://doi.org/10.1109/ACCESS.2020.3000006>
- [16] K. B. Adam, D. K. Silalahi, B. S. Aprillia, and H. A. Adhari, "Support vector regression method for predicting off-grid photovoltaic output power in the short term," *J. RESTI*, vol. 6, no. 4, pp. 548–554, 2022. <https://doi.org/10.29207/resti.v6i4.4134>
- [17] D. Markovics and M. J. Mayer, "Comparison of machine learning methods for photovoltaic power forecasting based on numerical weather prediction," *Renew. Sustain. Energy Rev.*, vol. 161, p. 112364, 2022. <https://doi.org/10.1016/j.rser.2022.112364>
- [18] A. Keddouda, R. Ihaddadene, A. Boukhari, A. Atia, M. Arıcı, N. Lebbihiat, and N. Ihaddadene, "Solar photovoltaic power prediction using artificial neural network and multiple regression considering ambient and operating conditions," *Energy Convers. Manag.*, vol. 288, p. 117186, 2023. <https://doi.org/10.1016/j.enconman.2023.117186>
- [19] M. J. Deka, P. Kalita, D. Das, A. D. Kamble, B. J. Bora, P. Sharma, and B. J. Medhi, "An approach towards building robust neural networks models using multilayer perceptron through experimentation on different photovoltaic thermal systems," *Energy Convers. Manag.*, vol. 292, p. 117395, 2023. <https://doi.org/10.1016/j.enconman.2023.117395>
- [20] R. Asghar, F. R. Fulginei, M. Quercio, and A. Mahrouch, "Artificial neural networks for photovoltaic power forecasting: A review of five promising models," *IEEE Access*, vol. 12, pp. 90 461–90 485, 2024. <https://doi.org/10.1109/ACCESS.2024.3420693>
- [21] M. Terkes, A. Demirci, I. Dagal, E. Gokalp, and U. Cali, "Comparative analysis of AI-based regression models for solar power forecasting: Techniques, preprocessing, and evaluation," *Energy Rep.*, vol. 15, p. 109071, 2026. <https://doi.org/10.1016/j.egyr.2026.109071>
- [22] M. Ali, A. Rabehi, A. Souahlia, M. Guermoui, A. Teta, I. E. Tibermacine, A. Rabehi, M. Benghanem, and T. F. Agajie, "Enhancing PV power forecasting through feature selection and artificial neural networks: A case study," *Sci. Rep.*, vol. 15, p. 22574, 2025. <https://doi.org/10.1038/s41598-025-07038-x>
- [23] A. Kumar, A. K. Dubey, I. S. Ramírez, A. M. del Río, and F. P. G. Márquez, "Artificial intelligence techniques for the photovoltaic system: A systematic review and analysis for evaluation and benchmarking," *Comput. Methods Eng.*, vol. 31, pp. 4429–4453, 2024. <https://doi.org/10.1007/s11831-024-10125-3>
- [24] M. Hammad, S. Khalil, and I. M. Mahmoud, "Application of nonparametric ML on forecasting the production of a large-scale solar power plant: Kom-Ombo case study," *Sustain. Comput. Informatics Syst.*, vol. 45, p. 101074, 2025. <https://doi.org/10.1016/j.suscom.2024.101074>
- [25] A. J. Lari, A. P. Sanfilippo, D. Bachour, and D. Perez-Astudillo, "Using machine learning algorithms to forecast solar energy power output," *Electronics*, vol. 14, no. 5, p. 866, 2025. <https://doi.org/10.3390/electronics14050866>
- [26] F. Aksan, V. Suresh, and P. Janik, "PV generation prediction using multilayer perceptron and data clustering for energy management support," *Energies*, vol. 18, no. 6, p. 1378, 2025. <https://doi.org/10.3390/en18061378>
- [27] I. M. Mfetoum, S. K. Ngoh, R. J. J. Molu, B. F. N. Kenfack, R. Onguene, S. R. D. Naoussi, J. G. Tamba, M. Bajaj, and M. Berhanu, "A multilayer perceptron neural network approach for optimizing solar irradiance forecasting in Central Africa with meteorological insights," *Sci. Rep.*, vol. 14, p. 3572, 2024. <https://doi.org/10.1038/s41598-024-54181-y>
- [28] P. Di Leo, A. Ciocia, G. Malgaroli, and F. Spertino, "Advancements and challenges in photovoltaic power

forecasting: A comprehensive review,” *Energies*, vol. 18, no. 8, p. 2108, 2025. <https://doi.org/10.3390/en18082108>

- [29] C. E. Ebeling, *An Introduction to Reliability and Maintainability Engineering*. Waveland Press, Inc., 2019.
- [30] A. Sabadus, R. Blaga, S. M. Hategan, D. Calinoiu, E. Paulescu, O. Mares, R. Boata, N. Stefu, M. Paulescu, and V. Badescu, “A cross-sectional survey of deterministic PV power forecasting: Progress and limitations in current approaches,” *Renew. Energy*, vol. 226, p. 120385, 2024. <https://doi.org/10.1016/j.renene.2024.120385>

Nomenclature

η_{PV}	Solar panel efficiency (%)
η_{SCC}	Solar charge controller efficiency (%)
P_{PV}	Output power of solar panel (W)
P_{SCC}	Output power of solar charger controller (W)
I_r	Solar Irradiance (W/m^2)
A_{PV}	Solar panel area
V	Voltage
I	Electric current
P	Electrical power
T	Temperature
SOC	State of charge
$R(i)$	Reliability at the i -th data point
N_D	Total number of failures

Greek symbols

η	Efficiency (%)
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Subscripts

PV	Photovoltaic
SCC	Solar charge controller
i	Data/time index
r	Irradiance